Superposition & Reflection

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Superposition & Reflection

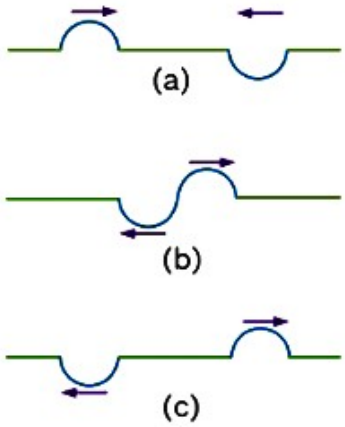
THE SUPERPOSITION PRINCIPLE

Disturbance produced by two or more sources passing through a point is the sum of disturbance produced by each wave. Writing above principle in mathematical form we get,

$$y(x, t) = y_1(x, t) + y_2(x, t)$$

In general this principle is only applicable to small disturbances. For large disturbance this principle is not valid and such waves are called non-linear wave.

The principle of superposition can also be expressed as a resultant wave produced by overlapping of the waves that algebraically adds in order to produce it. Also the overlapping waves do not in any way interfere the path of one other.



The resultant waveform is always the sum of wave functions of individual waves If we have two or more waves moving in that medium

Q **The expression of two waves passing through is given by**

$$y = (1.0\ cm)\ \sin[(3.14\ cm^{-1})x - (157\ s^{-1})t]$$

and $$y = (1.5\ cm)\ \sin[(1.57\ cm^{-1})x - (314\ s^{-1})t].$$

Find the net displacement of particle at position x = 4.5 cm and at time t = 5 ms,

Sol. As per the superposition principle, the resultant disturbance is a vector sum of individual disturbances.

$y_1(x = 4.5\ cm, t = 5\ ms) = (1.0\ cm)\ \sin[(3.14\ cm^{-1})(4.5\ cm) - (157\ s^{-1})(5.0 \times 10^{-3} s)]$

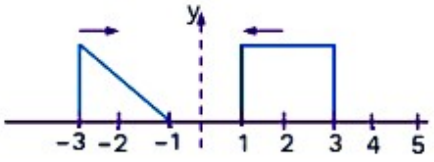
$$= (1.0\ cm)\ \sin\left[4.5\pi - \frac{\pi}{4}\right] = \frac{1.0\,cm}{\sqrt{2}}$$

and $y_2(x = 4.5, t = 5\ ms) = (1.5\ cm)\ \sin[(1.57\ cm^{-1})(4.5\ cm) - (314\ s^{-1})(5.0 \times 10^{-3}\ s)]$

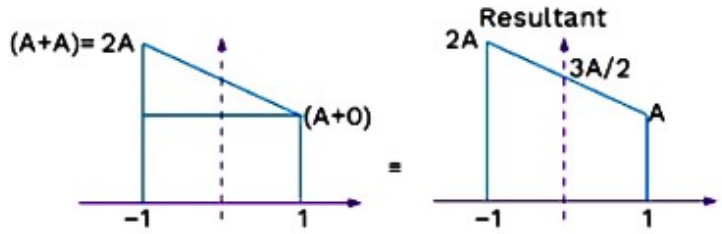
$$= (1.5\ cm)\ \sin\left[2.25\pi - \frac{\pi}{2}\right] = (1.5\ cm)\ \sin\frac{\pi}{4} = -\frac{1.5cm}{\sqrt{2}}$$

The net displacement is $y = y_1 + y_2 = \frac{-0.5\,cm}{\sqrt{2}} = -0.35\ cm.$

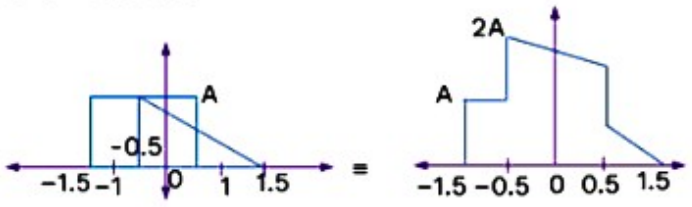
Q **Figure shows two waves travelling in opposite direction with same speed 1 cm/sec draw the shape of resultant wave at 2 sec and 2.5 sec respectively.**



Sol. At time t = 2 sec, first triangular wave pulse moves 2 cm forward and square wave pulse moves 2 cm in backward direction.



Similarly at time t = 2.5 sec



Interference of Waves Going in Same Direction

Assume that two identical sources send sine waves with the same angular frequency ω in the positive x direction. Wave speed would be same as they are produced in same medium but since sources are located at different points therefore, they will differ in phase. The wave function is represented as

$y_1 = A_1 \sin(kx - \omega t)$ and

$y_2 = A_2 \sin(kx - \omega t + \phi)$

According to the superposition principle, the resultant wave is given by

$y(x, t) = y_1(x, t) + y_2(x, t) = A_1 \sin(kx - \omega t) + A_2 \sin(kx - \omega t + \phi)$

we get $y = A \sin(kx - \omega t + \alpha)$

where, $A_{net} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos\phi}$ (A_{net} is amplitude of the resultant wave)

Also, $\tan\alpha = \dfrac{A_2 \sin\phi}{A_1 + A_2 \cos\phi}$

(α is phase difference of the resultant wave with the first wave).

Constructive and Destructive Interference:

Constructive Interference:

When resultant amplitude A_{net} is maximum

$$A_{net} = A_1 + A_2$$

When, $\cos\phi = +1$ or $\phi = 2n\pi$ ...here n is an integer.

Destructive Interference:

When resultant amplitude A_{net} is minimum

or $A_{net} = |A_1 - A_2|$

when $\cos\phi = -1$ or $\phi = (2n + 1)\pi$ ...where n is an integer.

Q **Two sine waves of the same frequency travel along a chord in the same direction. If A_1 = 3.0 cm along with A_2 = 4.0 cm, ϕ_1 = 0 and ϕ_2 = π/2 rad, Find the amplitude of the generated wave?**

Sol. Resultant amplitude = $\sqrt{3^2 + 4^2 + 2 \times 3 \times 4 \times \cos 90^\circ}$ = 5 cm.

Reflection and Transmission of Waves

Traveling waves on rigid or denser boundaries are reflected in antiphase, but no phase will change when reflections on open boundaries (rare media) occur. The transmitted wave will never invert, but the propagation constant k will change.

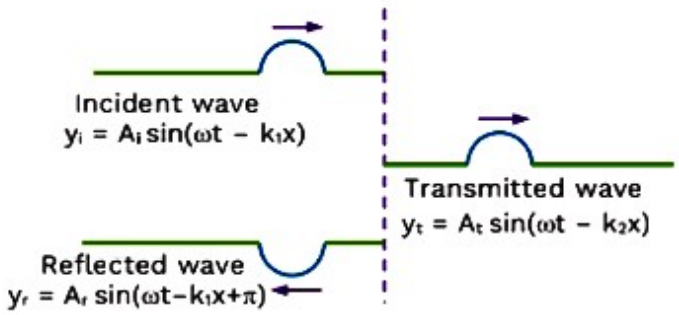


Fig: Reflection at denser boundary

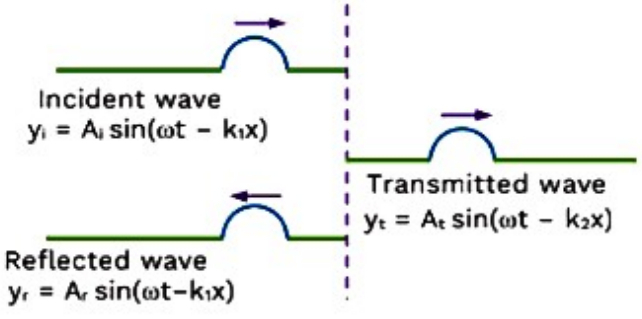
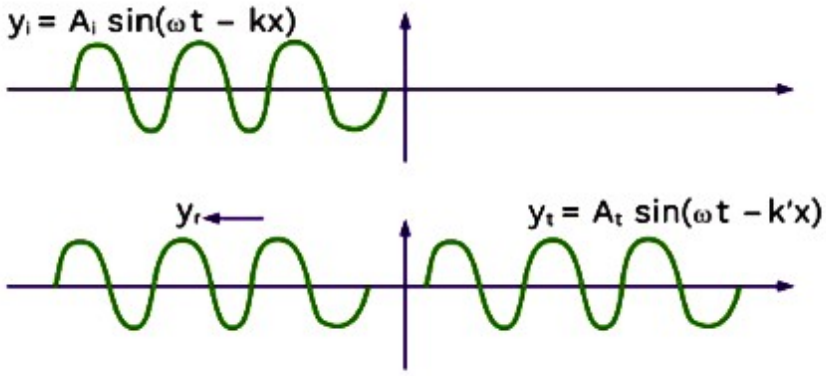


Fig: Reflection at rarer boundary

Reflection and Transmission of the Wave:

Whenever a wave suffers change of medium at interface, the wave is partially reflected and partially transmitted.

Let us assume incidence wave given by $y_i = A_i \sin(\omega t - kx)$ and reflected wave, $y_r = A_r \sin(\omega t + kx)$ and transmitted wave, $y_t = A_t \sin(\omega t - k'x)$



Frequency of the wave is property of source and it remains same both for reflected wave and transmitted wave.

Wavelength and wave speed of transmitted wave will change when wave goes from one medium to another.

Now, applying boundary condition at origin

At $x = 0$, $y_i + y_r = y_t$

$\Rightarrow A_i \sin\omega t + A_r \sin\omega t = A_t \sin\omega t$

$\therefore A_i + A_r = A_t$ ...(i)

Also,

$$P_i = P_r + P_t$$

$$= \frac{1}{2}\mu A_i^2\omega^2 v_1 = \frac{1}{2}\mu A_r^2\omega^2 v_1 + \frac{1}{2}\mu_2 A_t^2\omega^2 v_2$$

$$\Rightarrow \frac{T}{v_1^2}A_i^2 v_1 = \frac{T}{v_1^2}A_r^2 v_1 + \frac{T}{v_2^2}A_t^2 v_2$$

$$\Rightarrow \frac{A_i^2}{v_1} - \frac{A_r^2}{v_1} = \frac{A_t^2}{v_2}$$

Now,

$$\frac{(A_i + A_r)(A_i - A_r)}{v_1} = \frac{(A_t)^2}{v_2}$$

$$\Rightarrow \frac{A_i - A_r}{v_1} = \frac{A_t}{v_2}$$ [since $A_i + A_r = A_t$ from equation (i)]

$$\Rightarrow A_i - A_r = \frac{A_t v_1}{v_2}$$ ...(ii)

From equation (i) and (ii), we get

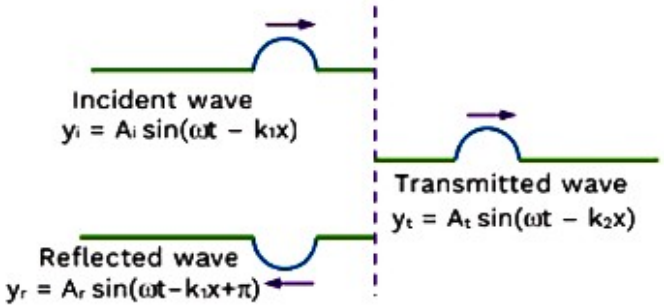
$$2A_i = A_t\left(\frac{v_1 + v_2}{v_2}\right)$$

$$\Rightarrow \quad A_t = \left(\frac{2v_2}{v_1 + v_2}\right)A_i, \; A_r = A_i\left(\frac{v_2 - v_1}{v_1 + v_2}\right)$$

Case 1: Reflection at Denser boundary

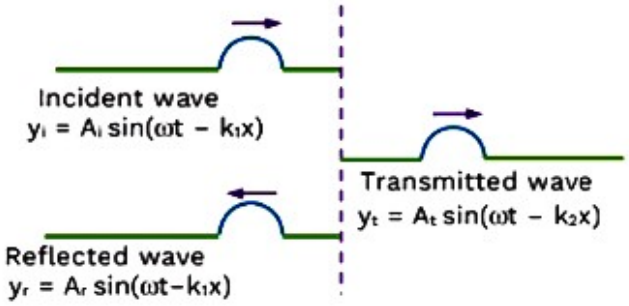
$\Rightarrow$ $\mu_2 > \mu_1$ $v_2 < v_1$, A_r & A_i will be of opposite sign.

Reflected wave will suffer phase change of π i.e., in opposite phase.



Case 2: Reflection at Rarer boundary

$\Rightarrow$ $\mu_2 < \mu_1$ $v_2 > v_1$, A_r & A_i will be of same sign.



Q **Wave of harmonics moving on string 1 get partially reflected and partially transmitted from point of intersection with string 2 having linear mass density four times that of string 1. If the expression of the incident wave is $y_i = A_i \cos(k_1 x - \omega_1 t)$. Write the expression of transmitted and the reflected waves in terms of A_i, k_1 and ω_1?**

Sol. Since $T_2 = T_1$ and $\mu_2 = 4\mu_1$

Using, $v = \sqrt{T/\mu}$,

We get, $v_2 = \frac{v_1}{2}$ ...(i)

Since frequency remains same means, $\omega_1 = \omega_2$ ...(ii)

Wave number, $k_2 = \frac{\omega_2}{v_2} = \frac{\omega_1}{v_1/2} = 2\frac{\omega_1}{v_1} = 2k_1$ ...(iii)

The amplitudes are,

$$A_t = \left(\frac{2v_2}{v_1 + v_2}\right)A_i = \left[\frac{2(v_1/2)}{v_1 + (v_1/2)}\right]A_i = \frac{2}{3}A_i \quad \text{...(iv)}$$

and $$A_r = \left(\frac{v_2 - v_1}{v_1 + v_2}\right)A_i = \left[\frac{(v_1/2) - v_1}{v_1 + (v_1/2)}\right]A_i = \frac{A_i}{3} \quad \text{...(v)}$$

Now by using above values we can write the equation of transmitted wave as,

$$\boxed{y_t = \frac{2}{3}A_i \cos(2k_1x - \omega_1 t)}$$

Similarly equation of reflected wave would be

$$\boxed{= \frac{A_i}{3}\cos(k_1x + \omega_1 t + \pi)}$$